Instawards Statement of Work (SOW)
30-Day Scoped Engagement

Network: Stellar testnet only; mainnet deployment is reserved for a later phase.

Scope: One protected Stellar x402 USDC agent payment flow only: MCP/HTTP request bound authorization, Soroban payment policy, deterministic velocity circuit breaker, and passkey-based signer freeze/revocation.

Regulation Posture: Non custodial, no yield, no real value transactions; Beaver402 operates only with testnet USDC and does not provide custody, investment, or compliance services.

Drafting Method: Drafted with an adversarial AI persona review method to stress test scope, risks, edge cases, and completion evidence; not presented as a multi person team.

1. Project & Team Information

Project Name:	Beaver402
Builder / Team Name:	Emir Yenikale
Primary Contact (Name + Email):	emiryenikale@gmail.com
Ambassador Chapter:	**Stellar Türkiye**
Ambassador Chapter Lead:	**İrem Koçi (Stellar Türkiye)**
Date Submitted:	17.07.2026
Suggested Sprint Start Date:	20.07.2026

2. Instawards Overview & Intent

2.1 Instawards Purpose (for Builder Context)

Instawards are designed to support short, clearly scoped, execution-focused work that helps a project make tangible progress toward building on Stellar. Instawards are meant to fund specific, achievable outcomes that can be completed and demonstrated within 30 days or less.

This SOW represents a shared commitment between the Builder and the Ambassador Chapter Lead on what will be delivered, why it matters, and how success will be verified.

3. Problem Statement & Objective

Problem Being Addressed	What specific problem, gap, or blocker is this	Stellar x402 enables AI agents to execute autonomous USDC payments, but a payment

	Instaward intended to solve?	that is valid under the protocol can still be inconsistent with the task approved by the account owner. A compromised prompt, unsafe tool, software error, or retry loop may change the endpoint, request content, recipient, or amount, or create many individually valid payments within a short period. Buyer side intent binding alone also leaves a trust gap. Soroban cannot read an HTTP request and cannot independently determine whether a buyer adapter accurately represented what the merchant received. A compromised request pipeline could commit to one description while sending a different request. Beaver402 addresses both gaps with a two party Proof of Intent. The merchant signs a canonical challenge describing the paid request and settlement terms, while the buyer creates an independently approved payment intent covering the same fields. A Soroban smart account policy authorizes settlement only when both signatures and all security critical fields match. The policy also prevents replay, enforces deterministic velocity limits, automatically freezes abnormal payment bursts, and allows the owner to revoke the delegated agent signer on chain using a passkey.
Objective of This Instaward	In one or two sentences, what will be true at the end of 30 days if this Instaward is successful?	At the end of 30 days, Beaver402 will deliver a working open source Stellar testnet MVP in which a merchant signed MCP or HTTP request challenge and a buyer approved payment intent must cryptographically match before an x402 USDC payment is authorized. The system will enforce replay and velocity rules through Soroban, emit a verifiable Proof of Intent event, and allow the account owner to freeze payments and revoke the delegated agent signer using passkey authentication.
Example prompts for builders: What is currently preventing progress? What is unclear, missing, or unbuilt today? Why is this problem worth solving now?		

4. Scope of Work (30-Day Deliverables)

Important guidance: This scope must be achievable within **30 calendar days**. If the work feels larger, it should be reduced or split into more achievable phases.

4.1 In-Scope Deliverables

Deliverable	Description (What will be built or produced?)	Why this matters
D1: Soroban Payment Policy	A Rust Soroban smart account policy will be deployed on Stellar testnet with a passkey owner and delegated agent signer. Before authorizing a USDC payment, the policy will verify the approved buyer signer, allowlisted merchant public key, merchant challenge signature, domain separated challenge and intent hashes, asset, recipient, amount, network, nonce, and expiry. The policy will require the merchant signed request digest and settlement terms to match the buyer approved intent. It will enforce single use challenge and intent protection, deterministic velocity limits, and automatic payment freezing when configured transaction count or amount thresholds are exceeded. Passkey protected owner functions will support freezing payments, restoring access, and revoking the agent signer. Contract events, explicit errors, and unit tests will cover valid authorization, invalid merchant signatures, merchant and buyer mismatches, replay, expiry, velocity violations, and unauthorized recovery actions. A successful authorization will emit a privacy preserving proof of intent event containing hashes and settlement identifiers rather than the HTTP body.	Neither the agent nor a compromised buyer side adapter can unilaterally redefine an approved paid request. The on chain policy enforces agreement between merchant terms and buyer authorization while preserving immediate owner control.

D2: Intent Bound Stellar x402 Payment Flow	A TypeScript reference flow will connect one MCP agent workflow, one demo merchant endpoint, and a Stellar testnet x402 USDC payment. The merchant will return a signed challenge containing a version and domain separator, merchant identity, HTTP method, normalized endpoint, request body hash, recipient, asset, amount, network, nonce, and expiry. The Beaver402 adapter will independently capture the MCP tool call and the same HTTP and settlement fields, then create a deterministic buyer payment intent. It will compare the merchant challenge with the approved request before requesting authorization. The Soroban policy will verify both sides and validate their shared fields against the actual USDC payment parameters before settlement. The delivery includes documented canonical encoding, domain separation rules, TypeScript and Rust cross language test vectors, and a reference merchant challenge signer. .	This creates a portable and testable security primitive. Payment is permitted only when the merchant's signed request and the buyer's approved intent describe the same action and settlement.
D3: Passkey Control and Verification	A minimal control interface will allow the account owner to authenticate with a passkey, view the delegated signer and policy state, inspect the merchant identity and Proof of Intent status for the demo flow, freeze payments, restore access, and revoke the agent signer through Stellar testnet transactions. The completed system will be tested against valid payments, body or endpoint mutations, recipient or amount mismatches, untrusted merchants, invalid merchant signatures, expired challenges or intents, replay attempts, abnormal payment velocity,	This makes the new security guarantee understandable and independently verifiable while retaining immediate human control during unsafe agent behavior.

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<tr><td></td><td>unauthorized recovery actions, frozen account behavior, and revoked signer access.

Delivery will include a public repository, deployment instructions, automated test results, contract addresses, transaction and event links, and documentation.</td><td></td></tr>
<tr><td colspan="3">Out of Scope (Explicitly Not Included)</td></tr>
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Mainnet deployment is not included. Beaver402 will operate exclusively on Stellar testnet using testnet USDC. Real value transactions are outside the scope.
Production security services are not included. This excludes an independent third party audit, production certification, KMS, HSM, institutional custody, and enterprise key management.
Merchant infrastructure beyond the reference implementation is not included. Beaver402 will not provide a global merchant registry, PKI, merchant onboarding network, production certificate lifecycle, or support for arbitrary third party merchants. One reference merchant signer will be delivered.
Beaver402 will not develop a new wallet, facilitator, payment network, or replacement for x402.
Raw HTTP request bodies and private MCP content will not be stored on chain. Only domain separated hashes and settlement identifiers will be recorded.
The intent and challenge schema, along with the test vectors, will serve as a reference specification. Formal standardization is reserved for a later phase.
The MVP will use deterministic velocity rules. Behavioral profiling, predictive models, and automated risk scoring are outside the scope.
Additional blockchains, assets, payment protocols, mobile applications, enterprise dashboards, compliance systems, advanced analytics, and production monitoring are not included.
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4.2 Deliverable-Aligned Budget Request

Requested Budget Amount	Rationale for Budget Request
$5,000	This budget covers 30 days of development, integration, testing, and public delivery of Beaver402. The requested $5,000 is allocated across the three in-scope deliverables based on development effort, integration complexity, and technical risk. Soroban Proof-of-Intent Policy - $2,000 Design, implementation, testing, and Stellar testnet deployment of the Rust smart account policy. This includes merchant and buyer signature verification, payment-field matching, replay protection, nonce and expiry validation, velocity limits, automatic freezing, passkey-protected owner actions, signer revocation, contract events, and error handling. Two-Party Proof-of-Intent x402 Flow - $2,000 Development of the merchant challenge signer, MCP/x402 adapter, buyer intent generator, canonical encoding rules, shared TypeScript/Rust test vectors, and integration with the Soroban policy, x402 facilitator, and testnet USDC settlement. Passkey Control, Security Validation, and Final Delivery - $1,000 Development of the owner interface for viewing policy status, freezing and restoring payments, and revoking the delegated signer. This also covers adversarial testing, bug fixes, documentation, automated test results, transaction evidence, and the final demo video.

	Total: $5,000 The final result will be a public and independently verifiable Stellar testnet MVP. Mainnet deployment, third-party audits, production infrastructure, formal standardization, and broader merchant support are outside this scope.

5. 30-Day Execution Plan & Timeline

5.1 Weekly Breakdown

Week	Planned Work	Expected Output
Week 1: Validate the x402 Smart Account Payment Path	Establish the repository, development environment, Stellar testnet accounts, testnet USDC balances, x402 facilitator connection, and one paid reference endpoint. Create a smart account with a passkey owner and delegated agent signer. Define the versioned and domain separated merchant challenge and buyer intent, including field normalization, hash algorithm, signature format, privacy boundary, nonce, and expiry. Produce shared TypeScript and Rust positive and negative test vectors. Validate one baseline x402 USDC payment and confirm how Soroban authorization entries bind to settlement.	One successful baseline payment with a transaction hash, a documented threat model and architecture, a canonical Proof of Intent specification, and matching TypeScript and Rust test vectors.

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Week 2: Build and Deploy the Soroban Payment Policy	Develop the Rust Soroban policy. Implement approved buyer signer and merchant key checks, merchant signature verification, challenge and intent field matching, payment parameter validation, nonce and expiry handling, replay protection, velocity tracking, automatic freezing, passkey protected recovery, signer revocation, explicit errors, and privacy preserving Proof of Intent events. Add contract and cross language tests.	An optimized policy deployed on Stellar testnet with a verifiable address. Tests demonstrate valid two party authorization and rejection of invalid signatures, mismatches, expiry, replay, velocity violations, and unauthorized owner actions.
Week 3: Complete the Protected Agent Payment Flow	Integrate the reference merchant challenge signer, MCP adapter, buyer intent generator, Soroban policy, x402 client, testnet USDC settlement, and facilitator. Keep the signing key outside the language model. Build a minimal passkey interface for signer and policy status, merchant identity, Proof of Intent result, freeze, restore, and revoke actions. Add transaction confirmation and clear rejection messages.	A functional MVP in which a valid merchant challenge and buyer intent complete an x402 payment. Mutating the request body, endpoint, recipient, amount, asset, or network prevents authorization. Abnormal velocity freezes payments, and passkey revocation blocks the agent.
Week 4: Security Validation, User Testing, and Final Delivery	Execute adversarial tests for request mutation, settlement mismatch, untrusted merchants, invalid signatures, expired challenges or intents, replay attempts, abnormal velocity, unauthorized recovery, frozen states, and revoked signers. Review authorization, storage,	A public open source repository with passing tests, deployed testnet contracts, transaction and event evidence, an adversarial test report, and a video showing a valid Proof of Intent payment, mutation rejection, automatic circuit breaker activation, and passkey based signer revocation.

	amount boundaries, resource usage, event privacy, and error handling. Resolve identified issues and complete the public repository, reproducible setup and deployment guide, security assumptions, known limitations, test evidence, and demonstration.	

6. Evidence of Completion (Required)

Important guidance: Evidence should be clear, verifiable, and easy to review by the Ambassador Chapter Lead **with minimal technical expertise**.

6.1 Planned Evidence to Be Submitted

Deliverable	**Evidence Type** (link, repo, demo, screenshot, doc, tx hash, etc.)	Description
D1: Soroban Payment Policy	Public repository, specification, test vectors, testnet contract link, and automated test report.	Open the testnet link to verify the policy contract and its Proof of Intent and policy events. Review tests confirming buyer and merchant signatures, field matching, payment parameters, expiry, replay protection, velocity limits, automatic freezing, and owner function protection. Run the shared TypeScript and Rust vectors to confirm deterministic encoding and hashing.
D2: Protected x402	Transaction and event links, public repository, reference merchant endpoint, and demonstration video.	Verify a successful testnet USDC payment and its Proof of Intent event. Watch the merchant issue a signed

Payment Flow		challenge, the buyer create a matching intent, and the policy authorize settlement. Observe rejection when the request body, endpoint, recipient, amount, asset, network, or merchant signature is modified, as well as when a challenge is replayed.
D3: Passkey Control and Final Verification	Transaction links, demonstration video, screenshots, and adversarial test report.	Watch the owner authenticate with a passkey, inspect the policy and Proof of Intent state, freeze and restore payments, and revoke the delegated signer. Verify on chain control actions and observe that a revoked signer cannot authorize later payments. Review the complete security scenario report.

6.2 Evidence Verification Checklist (For Ambassador Use)

For each deliverable, the Ambassador Chapter Lead will assess whether evidence is present and sufficient.

Deliverable	Evidence Present	Evidence Partial	Evidence Missing	Comments
Deliverable 1	☐	☐	☐	
Deliverable 2	☐	☐	☐	
Deliverable 3	☐	☐	☐	